

Pranavsai Gandikota

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EDUCATION

University of Central Florida

Orlando, FL

B.S. Computer Science | **GPA: 3.99/4.0** | President's Honor Roll 2024, 2025 | Dean's List 2026

Aug. 2024 – May 2028

Activities: SASE (CS Tech Chair), AI UCF (Projects Director), KnightHacks (Web Dev/Design), SHPE

EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

Ford Motor Company (Global Trade Services, Order Fulfillment, Marketing & Sales Technology)

Dearborn, MI

- Cut processing time by **92%** (45 to 3.5 minutes) by migrating **6 Java Struts batch jobs** to **Java 21/Spring Boot** on **GCP Cloud Run**, shifting data-heavy work into **PostgreSQL stored procedures**.
- Automated scheduled fixed-width file generation and **SFTP** delivery to **GECHUB** for QAD processing; validated **200K+ records** against legacy outputs to ensure formatting and integration without affecting downstream processes.
- Added **143 automated tests** with **90%+ coverage** and resolved **31 critical SonarQube findings**, including SQL injection and credential-logging risks, before release through **Tekton CI/CD**.
- Built automated secret refresh using **Secret Manager, Pub/Sub, Cloud Functions, and Spring Actuator**, reloading affected GOaLS services in **under 2 minutes** without restarts; provisioned with **Terraform**.

Teaching Assistant – Object-Oriented Programming with Java

Jan. 2026 – May 2026

College of Engineering and Computer Science, UCF

Orlando, FL

- Graded assignments and exams for correctness and algorithmic efficiency, and held weekly office hours supporting **166 students** on **Java OOP** design, programming, and debugging.

Research Assistant

Sep. 2024 – Dec. 2025

Interactive Systems and User Experience (ISUE) Lab, UCF

Orlando, FL

- Fine-tuned **3 Llama 3.1 8B adapters** on **1,820 NASA-TLX and SUS examples** using **PyTorch, CUDA, Hugging Face Transformers, and QLoRA** with 4-bit quantization.
- Built a **Python** text-to-3D workflow spanning scene extraction, **Stable Diffusion 3.5**, human review, and **TRELLIS**; processed **135 images across 30 scene directories** and metadata for **48 objects**.
- Orchestrated long-running GPU jobs with subprocesses, thread-safe queues, watchdog timers, and isolated Conda environments while keeping the review interface responsive.

PROJECTS

Paradise | Node.js, WebRTC, ONNX Runtime, Depth Anything V2

Sep. 2026

- Won **3rd Best Overall** at ShellHacks 2026 among **290+ projects** and **1,400+ hackers**.
- Built touch-only navigation for DeafBlind users by streaming phone video over **WebRTC** to a laptop perception pipeline and converting object, obstacle, GPS, and compass cues into Joy-Con haptics.
- Kept obstacle guidance responsive by running **Depth Anything V2** separately from slower object detection, estimating approximate metric distances at **80ms/frame on the laptop** with a **RANSAC** floor-plane fit.

uKnight | Java, Spring Boot, Redis, WebRTC, WebSockets, Next.js, PostgreSQL, GCP, Docker

Feb. 2026 – Jun. 2026

- Led a **6-member team** building an anonymous university video-chat platform; deployed the **Spring Boot/WebSocket** backend on **GCP Cloud Run** for WebRTC signaling, reaching **72 verified users in 3 weeks**.
- Moved matchmaking and session state to a distributed **Redis ZSET**, using an atomic **Lua** operation to prevent double-matches and stale-session pruning to support multiple backend instances.

UCF SASE Official Website | Next.js, TypeScript, Supabase, PostgreSQL, Vercel

Jul. 2026 – Present

- Led the web team as CS Tech Chair to replace manual attendance tracking with a **Next.js/Supabase** QR check-in system for **200+ members**, with RLS policies and duplicate-check-in protection.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, C#, SQL

Frameworks & Web: Spring Boot, Spring Actuator, React, Next.js, Node.js, Express, WebRTC, WebSockets

Databases & APIs: PostgreSQL, MySQL, Redis, Supabase, REST APIs, OAuth 2.0

Cloud & DevOps: GCP (Cloud Run, Pub/Sub, Cloud Functions, Cloud SQL, Cloud Storage, Secret Manager), Terraform, Docker, Tekton, SonarQube, Linux

AI / ML: PyTorch, CUDA, Hugging Face Transformers, QLoRA, ONNX Runtime, Stable Diffusion 3.5, TRELLIS

Tools: Git, GitHub, Jira, Unity, Blender